

Multimodal Neurosymbolic Approach for Explainable Deepfake Detection

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Deepfake detection has become increasingly important in recent years owing to the widespread availability of deepfake generation technologies. Existing deepfake detection methods present two primary limitations; i.e., they are trained on a specific type of deepfake dataset, which renders them vulnerable to unseen deepfakes, and they regard deepfakes as a “black box” with limited explainability, making it difficult for non-AI experts to understand and trust the decisions. Hence, this article proposes a novel neurosymbolic deepfake detection framework that exploits the fact that human emotions cannot be imitated easily owing to their complex nature. We argue that deep fakes typically exhibit inter- or intra-modality inconsistencies in the emotional expressions of the person being manipulated. Thus, the proposed framework performs inter- and intra-modality reasoning on emotions extracted from audio and visual modalities using a psychological and arousal-valence model for deepfake detection. In addition to fake detection, the proposed framework provides textual explanations for its decisions. The results obtained using the Presidential Deepfakes Dataset and World Leaders Dataset of real and manipulated videos demonstrate the effectiveness of our approach in detecting deepfakes and highlight the potential of a neurosymbolic approach for expandability.

CCS Concepts: • **Social and professional topics** → **Social engineering attacks**; • **Computing methodologies** → **Scene anomaly detection**; **Biometrics**;

Additional Key Words and Phrases: Deepfake detection, Emotion recognition, Knowledge base, Manipulation detection, Neurosymbolic

This material is based upon work supported by the National Science Foundation (NSF) under award numbers 2329858 and 2231619, and Michigan Transnational Research and Commercialization (MTRAC), Advanced Computing Technologies (ACT) award number 292883. Any opinions, findings, conclusions, or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the NSF and MTRAC ACT.

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1551-6857/2024/09-ART341 \$15.00

<https://doi.org/10.1145/3624748>

ACM Reference format:

Ijaz Ul Haq, Khalid Mahmood Malik, and Khan Muhammad. 2024. Multimodal Neurosymbolic Approach for Explainable Deepfake Detection. *ACM Trans. Multimedia Comput. Commun. Appl.* 20, 11, Article 341 (September 2024), 16 pages.

<https://doi.org/10.1145/3624748>

1 INTRODUCTION

The advent of deep learning has resulted in the development of sophisticated tools for image and video manipulation. These tools can be used for legitimate purposes, such as for creating specific effects in the movie industry. However, they can be used by malicious parties, e.g., to create deepfakes. Deepfakes pose a significant threat to public trust and security, as they can be used to disseminate wrong information, defame individuals, or manipulate public opinion, specifically on social media [1]. For instance, a deepfake video can be created to show a political figure presenting false statements or engaging in illegal activities. An example in recent years includes a deepfake video of Canadian Prime Minister Justin Trudeau circulating on social media in 2019, which showed Trudeau discussing a scandal involving the Canadian government. Similarly, a deepfake video of former Vice President Joe Biden was created and shared during the 2020 election, when Biden appeared to endorse President Trump. These examples may result in public outrage over and loss of trust in an individual or institution; hence, the potential of deepfakes in influencing political campaigns or election outcomes is concerning. Deepfakes are not limited to politics [2]. In 2019, a UK-based energy company received a phone call from its CEO, which was in fact a deepfake voice generated by an AI algorithm, and was asked to transfer \$243,000 to a Hungarian supplier within an hour.

Researchers are developing various deep-learning-based methods to combat the dissemination of deep fakes. These methods focus primarily on visual artifacts that occur because of inconsistencies in the local texture. However, with the rapid advancement in deepfake technology, generating highly realistic content has become possible, thus rendering it increasingly challenging to differentiate between real and fake media using existing techniques. In other methods, biometric signals such as heartbeat rate [3], emotions [4], and eye-blinking [5] patterns are used, where the targeted facial motion of an individual is prioritized, thus limiting generalization to new identities. Furthermore, most deepfake detection techniques rely on black-box models that cannot interpret or explain the reason that a video is classified as fake. Hence, they do not garner the trust of the audience. Some explainable deepfake detection approaches have been reported in the literature; however, most of them perform decisions based on typical attention mechanisms such as Grad-CAM [6], that can only highlight a specific region via the heatmaps of an image or video. Despite the significant progress in deepfake detection, several research questions are yet to be addressed, as follows:

- (1) How can logical reasoning be effectively incorporated across multiple modalities to enhance deepfake detection?
- (2) What psychological knowledge is necessary to improve the generalizability of deepfake detection techniques?
- (3) How can explainable approaches be developed to make deepfake detection accessible to non-AI experts?

To answer these questions, this article proposes a multimodal neurosymbolic approach for deepfake detection. We argue that human emotion psychology is one of the most prominent biological signals for deepfake detection due to its complex nature, which cannot be perfectly replicated by generative algorithms. Therefore, we studied the specific probabilities of transitions from one

emotional state to another that results in consistent emotional stability in individuals. Our aim is to detect irregular transitions (with low probability) among various emotions in a single modality. However, we observed a natural connection between visual and aural emotions at a specified timestamp, which must be consistent in most cases. Therefore, the proposed method integrates multimodal and neurosymbolic approaches to perform human-understandable reasoning for deepfake detection. According to Henry Kautz [7], the proposed framework can be categorized into a Type 2 (i.e., Symbolic[Neuro]) architecture, where neural modules are used as subroutine within a symbolic problem solver. In summary, the symbolic and neural components are loosely coupled. Specifically, the proposed framework first extracts emotions from visual and aural modalities via deep models, which is the neural component. Subsequently, inter- and intra-modality reasoning is performed in the symbolic module to determine the fakeness of a specified video. Unlike existing explainable methods that use Grad-CAM [6] to highlight the portion of a frame to be focused, the proposed neurosymbolic approach can provide interpretable reasoning details that can be converted to textual explanations for non-AI experts. The main contributions of this study are summarized as follows:

- (1) A novel biological signal (which demonstrates consistency between emotion transitions) is proposed for deepfake detection. The proposed framework performs inter- and intra-modality reasoning to detect inconsistencies in emotional transitions in visual and aural modalities.
- (2) To the best of our knowledge, this study is the first to utilize external knowledge (i.e., psychological and arousal-valence dimensions) for deepfake detection using a neurosymbolic approach, which renders the overall framework more interpretable.
- (3) The proposed framework provides textual explanations (domain specific) with localization of the timestamp along with fakeness detection, which renders it more reliable and understandable for explaining decisions compared with state-of-the-art deepfake detection methods.

The remainder of the article is organized as follows: Section 2 presents the literature on deepfake detection algorithms. Section 3 presents the technical details pertaining to the proposed deepfake detection algorithm. The experimental settings and results are discussed in Section 4. Finally, the findings of the study and future research directions are presented in Section 5.

2 LITERATURE REVIEW

The literature on deepfake detection can be broadly classified into two categories: Section 2.1 artifact-based methods and Section 2.2 biological-signal-based methods. Artifact-based methods focus primarily on detecting blending traces that occur during deepfake generation, whereas biological-signal-based methods focus on individual behaviors and various psychological cues. Both categories consider spatial and temporal features along with single or multimodal data. The following subsections discuss each category in detail.

2.1 Artifact-based Methods

One of the most challenging aspects of deepfake generation algorithms is blending the source face with the target with minimum blending artifacts. To detect such artifacts for deepfake detection, Durall et al. [8] performed a discrete Fourier transform with an azimuthal average to analyze these artifacts in the frequency domain and make them more visible in the detection algorithm. Subsequently, several classification algorithms, such as the SVM, LR, and K-mean, were trained to classify deepfakes. Li and Lyu [31] combined VGG16, ResNet50, ResNet101, and ResNet152 to analyze artifacts in low-resolution deepfakes wrapped to fit the target face. This method was further

improved in another study [9] by adding a pooling layer, which resulted in better performance in detecting variations in the resolution of the target and deepfake media. Similarly, Nirkin et al. [10] first segmented a test image into face and hair/context and then generated three different encodings. Subsequently, these encodings were used in a decoder for classification, where the encoding of the source was concatenated with the difference between the face-only and hair/context encodings.

Masi et al. [11] proposed a two-stream framework consisting of bidirectional LSTM, with emphasis on residual errors for deepfake detection. One stream uses frequency to reduce the errors, whereas the other stream fine-tunes the original information in the color domain. Recently, Zhao et al. [12] proposed a transformer-based self-attention and self-subtracting mechanism for spatial artifact and temporal inconsistency detection. They mainly focused on investigating the supported learning mechanism by analyzing the proposed framework using the Grad-CAM [6]. Typical Deep-Fake videos often indicate inconsistencies in certain facial features across frames, which can be detected via appearance-level analysis. To detect such inconsistencies, Ge et al. [13] proposed a predictive representative learning approach that combines a CNN encoder, a ConvGRU-based aggregator, and a single-layer binary classifier to capture semantic changes specific to deepfakes. Some researchers have addressed deepfake detection as an anomaly detection problem. For example, Sun et al. [14] analyzed facial region trajectories for deepfake detection. In their method, the face region displacement was first calculated; subsequently, a trajectory series was constructed, which was then analyzed for deepfake detection.

2.2 Biological-signal-based Methods

Unlike GANs, which use sets of latent spaces for deepfake generation, recent diffusion probabilistic models have achieved high-quality deepfake generation with almost no residual artifact traces. Thus, biological signals such as heart rate, respiration, and emotions are considered better options for high-quality deepfake detection where visual artifacts are less apparent. Various techniques have been described in the literature that utilize heart rate analysis to detect deepfakes based on biological signals [15–17]. The challenge with these approaches is extracting accurate heart rates from visual signals. For instance, Jin et al. [18] used a multidimensional feature space from the RGB, YUV, and chromatic dimensions to extract heart rate from a visual modality. Subsequently, these multidimensional features were used to classify deep fakes using a CNN as a binary classifier. Similarly, Fernandes et al. [19] proposed a neural ordinary differential equation-based model for detecting heart rate changes over time. They extracted the heart rate by measuring skin color variations, optical intensities from the forehead, and temporal changes in color.

In addition to the heart rate, several individual behavior-based approaches have been reported for deepfake detection [20]. For example, Agarwal and Farid [21] investigated the dynamics of synchrony between audio and visual modalities. They argued that the movement of the mandible (lower jaw) caused changes in the shape of the ear and ear canal during talking. They analyzed the correlation between the corresponding facial landmarks and speech signals. Recently, Agarwal et al. [22] presented a similar approach, with emphasis on the synchronization between spoken words and facial movements in a video. They used pretrained facial landmark detection and a text-based model to extract facial movements and spoken words and learned person-specific biometric patterns.

Additionally, the potential of emotions in deepfake detection has been investigated. For example, Mazaheri and Roy-Chowdhury [23] customized a pretrained **facial expression recognition (FER)** model for deepfake detection with the additional application of localizing the section of the face that is involved in false expressions using Grad-CAM [6]. Mittal et al. [4] proposed a multimodal deepfake detection model that uses facial expression analysis and **speech emotion**

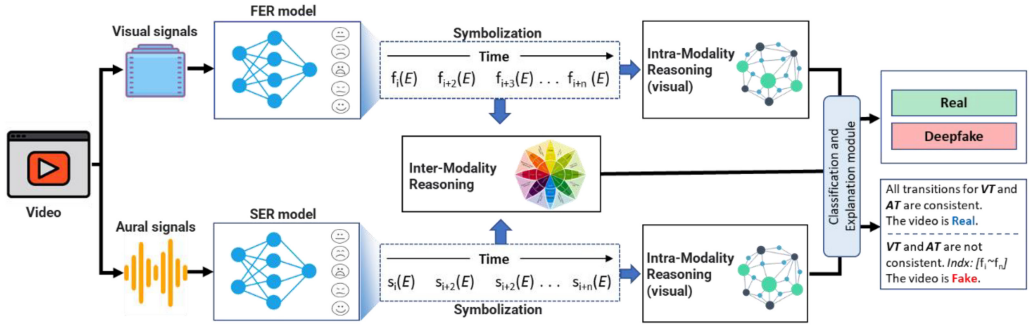


Fig. 1. Systematic overview of proposed deepfake detection framework, which comprises emotion detection and reasoning modules.

recognition (SER) to identify inconsistencies between the emotions portrayed in visual and audio channels, which are likely to occur in deepfake videos. Similarly, Hosler et al. [24] focused on the inconsistencies between aural and visual emotions for deepfake detection. In this method, the low-level descriptors of a subject's speech and face are first extracted to present the subject's emotions over time; subsequently, a recurrent neural network is trained on deepfake videos. These methods are closely related to the proposed method in terms of emotion analysis from multiple modalities. However, our study differs from [4, 23, 24] in two respects: we consider both inter- and intra-modality inconsistencies in emotions and perform reasoning based on psychological knowledge instead of deciding the fakeness of a video based on the error score.

3 PROPOSED METHODOLOGY

This section discusses the technical details of the proposed deepfake detection framework. The overall framework is shown in Figure 1. First, the input video is preprocessed and forwarded to the SER and FER models. Second, the predicted emotions are symbolized in the reasoning module, where intra- and inter-modality reasoning is performed to determine the fakeness of a given video. Finally, the framework predicts the fakeness of the video and provides a short textual explanation for interpretability. Each step in the proposed framework is discussed in the following subsections.

3.1 Problem Formulation

Consider V and A as the visual and aural modalities of the input video I to be tested for fakeness, respectively, whereas s_i and f_i are the audio segments and video frames of the synchronized timestamps that are preprocessed and forwarded to SER and FER models, respectively. The predicted emotions from V and A at timestamp i are denoted as $f_i(E)$ and $s_i(E)$, respectively. Our goal is to determine the fakeness of an input video with an explanation for which inter- and intra-modality symbolic reasoning is performed to detect fake emotion transitions based on the psychological knowledge base KB and arousal-valence dimensions AVD . The proposed framework can identify deepfake videos along with textual explanations that localize the timestamp and identify fakeness in inter- or intra-modalities. Table 1 presents the symbols and their descriptions. Algorithm 1 explains the overall flow of the proposed framework.

3.2 Preprocessing

The preprocessing step involves extracting V and A modalities from the input video, synchronizing and segmenting the aural signals into equal time intervals, and detecting the face in the selected frames. For synchronization, we selected three frames and partitioned the audio into three

Table 1. List of Acronyms Used in Proposed Framework

Symbol	Description	Symbol	Description
I	Input video	SER	Speech emotion recognition model
FER	Facial expression recognition model	s_i	Audio segment at time i
f_i	Video frame at time i	$f_i(E)$	Predicted emotion at f_i
$s_i(E)$	Predicted emotion at s_i	E	Set of all emotions
KB	Knowledge base	VT_i	Resultant visual emotion transition at time i
AT_i	Resultant aural emotion transition at time i	t	Threshold
AVD	Arousal-valence dimensions	VE	Set of detected visual emotions
AE	Set of detected aural emotions	IM	Resultant inter-modality reasoning

corresponding segments of a 1 s video to ensure that the audio and visual signals matched perfectly. Each audio segment lasted 0.5 s, with a 0.25 s overlap having mono channel at a sample rate of 16 kHz. Finally, we employed a pretrained face detector [25] on the selected frames to crop the face regions over time.

ALGORITHM 1: Emotion-inconsistency-based deepfake detection

Input: Video I of length n

Initialization: Load trained models and knowledge base, i.e., FER , SER , KB , AVM

Steps:

- 1: $f_i, s_i \leftarrow$ Preprocessing (I) // frames and segments extraction and synchronization
 - 2: **for** i to n **do**: // audio and visual emotion detection
 - $VE \leftarrow FER(f_i)$
 - $AE \leftarrow SER(s_i)$
 - 3: $VT, AT \leftarrow KB(VE, AE)$ // intra-modality reasoning using Equations (4) and (5)
 - 4: $IM \leftarrow AVD(VE, AE)$ // inter-modality reasoning using Equation (6)
 - 5: $pred, exp \leftarrow (VT, AT, IM)$ // classification and explanation generation using Equations (7) and (8)
-

Output: $pred, exp$

3.3 Multimodal Emotion Detection

Emotion is a prominent yet complex biological signal for the generation of deepfakes. Therefore, the preprocessed signals were forwarded to detect emotions over time from both the V and A of the input video. Facial landmarks, color, texture, and various acoustic properties such as pitch, intensity, and spectral attributes were considered. To extract emotions from both modalities, we employed pretrained FER and SER models that classified the input signals into seven basic emotions: angry, disgust, fear, happy, sad, surprise, and neutral.

3.3.1 Facial Emotion Recognition (FER). FER involves the identification and extraction of various facial landmarks such as the eyes, nose, mouth, and eyebrows. Subsequently, these landmarks are used to compute the movement of each landmark, thus providing valuable information regarding the emotional state of the individual. In the proposed framework, we adopted an FER model [26] based on the ResNet50 architecture. This model comprised multiple convolutional layers with ReLU activation, followed by max pooling and batch normalization. The output of the CNN was fed to a fully connected layer, which yielded the final classification via a softmax classifier. We trained this model on the FER2013 dataset [27] at a learning rate of 0.001 and a batch size of 128 for 50 epochs. Further technical details are available in an online repository [26].

Table 2. Knowledge Base of Emotional State Transition Acquired from Psychological Study [30]

	Normal	Happy	Sad	Angry	Disgust	Fear	Surprise
Normal	0.421	0.213	0.084	0.190	0.056	0.050	0.047
Sad	0.362	0.509	0.296	0.264	0.262	0.244	0.252
Happy	0.061	0.090	0.320	0.091	0.123	0.137	0.092
Angry	0.060	0.055	0.058	0.243	0.075	0.101	0.056
Disgust	0.027	0.039	0.108	0.086	0.293	0.096	0.164
Fear	0.034	0.051	0.064	0.076	0.069	0.279	0.075
Surprise	0.032	0.042	0.068	0.048	0.121	0.092	0.313

3.3.2 Speech Emotion Recognition (SER). To extract speech emotions from the preprocessed aural signals, we fine-tuned one of the existing implementations of the SER repository based on our problem. We retrained this model on seven basic emotions from the RAVDESS dataset [28] by excluding the “clam” class. First, the MFCC, Chroma, and MEL spectrograms were extracted; subsequently, they were fed into the MLP classifier. The best performance was achieved when the following parameters were used: a hyperparameter of $\alpha = 0.01$, a batch size of 256, an epsilon of $1e^{-08}$, and 300 hidden layers for 500 epochs. The dataset ratio was the standard 80% training set and 20% test set. Further technical details are available in [29].

3.4 Symbolic Reasoning Module

We argue that emotions are psychological phenomena that are not perfectly emulated during deepfake generation due to their complex nature. Therefore, the proposed symbolic reasoning module performs reasoning regarding the extracted emotions in two manners: (1) intra-modality reasoning, in which the inconsistency in detected emotions focuses on each modality over time, and (2) inter-modality reasoning, which focuses on inconsistencies in the valence-arousal space between aural and visual modalities. Intra- and inter-modality reasoning are based on the psychological knowledge base *KB* for emotional state transition and the valence-arousal distributions *VAD*, respectively. To perform logical reasoning, we first convert the detected aural and visual emotions into symbols over time, as shown in Equations (1) and (2):

$$VE = [f_i(E), f_{i+1}(E), f_{i+2}(E) \dots f_{i+n}(E)] \quad (1)$$

$$AE = [s_i(E), s_{i+1}(E), s_{i+2}(E) \dots s_{i+n}(E)], \quad (2)$$

where *VE* and *AE* are the sets of emotions detected from *V* and *A* modalities over time, respectively.

3.4.1 Intra-modality Reasoning. To detect inconsistencies between emotion transitions over time in each modality, we adopted the *KB* from a psychological study [30] to analyze human emotions in terms of the probability of emotional change from one state to another. This *KB* was acquired from 200 participants based on a questionnaire that represented the probability of transition between the current and next emotional states. We formulated this *KB* based on our problem and generated rules for reasoning. Table 2 presents the *KB*, where the first column represents the current emotional state and the first row represents the next detected emotional state. First, we calculated the mean value of all the transition probabilities of the detected emotion to set the threshold *t*, as expressed in Equation (3):

$$t = \text{mean}(KB(f_i(E) \mid s_i(E), \cdot)), \quad (3)$$

where *t* defines a unique threshold value for the transition probabilities of the seven emotions. Thus, we define a rule where if the probability of two constitutive emotional states exceeds *t*, then

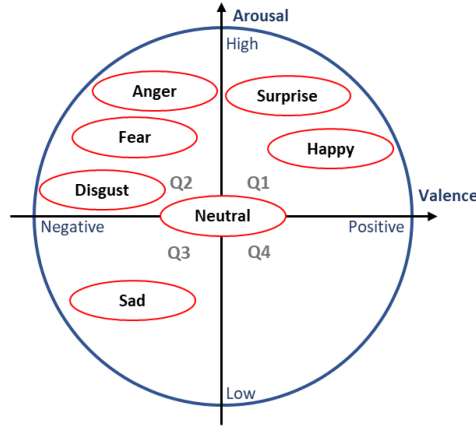


Fig. 2. Representation of seven basic emotions in arousal-valence dimension.

the transition will be considered normal (real); otherwise, the emotion transition will be considered unusual (fake), as expressed in Equations (4) and (5):

$$VT_i = \begin{cases} \text{real} & \text{if } KB(f_i(E), f_{i+1}(E)) > t \\ \text{fake} & \text{otherwise} \end{cases} \quad (4)$$

$$AT_i = \begin{cases} \text{real} & \text{if } KB(s_i(E), s_{i+1}(E)) > t \\ \text{fake} & \text{otherwise} \end{cases}, \quad (5)$$

where VT_i and AT_i are the emotion transition statuses for the visual and aural modalities at timestamp i , respectively.

3.4.2 Inter-modality Reasoning. The visual and aural modalities for deepfake detection have been compared previously [4, 31]. However, these methods rely on the final loss value to detect the overall fakeness of a given video, which is a black-box decision that does not specify the fake modality. Therefore, the proposed inter-modality reasoning module uses one of the well-established emotion models [32] (i.e., arousal-valence dimensions), which is typically used in psychology, and captures the emotional affect through valence and the intensity of the emotion through arousal. For this task, we formulated the basic principles of the model for seven basic emotions to accommodate our problem. We distributed the detected emotions into four quadrants, Q_n , where $n = 1, 2, 3$, and 4 , as shown in Figure 2. To define the rule for determining the variation in emotions in visual and aural modalities, a video is considered real if both aural and visual emotions belong to the same quadrant Q_n at a time i , and vice versa. For example, if the audio modality detects the happy emotion and the visual modality detects the surprise emotion, then the result will be a typical (real) video because both belong to the same quadrant Q_1 . The rule for this task is formulated as shown in Equation (6):

$$IM_i = \begin{cases} \text{real} & \text{if } f_i(E) \wedge s_i(E) \in Q_n \\ \text{fake} & \text{otherwise} \end{cases}, \quad (6)$$

where IM_i is the resultant status of inter-modality reasoning for emotions V and A detected at time i , and Q_n is one of the quadrants where $n \in [1, 2, 3, 4]$, as shown in Figure 2.

3.5 Classification and Textual Explanation Generation

Once intra- and inter-modality reasoning is performed, the proposed framework generates a short textual explanation with probability scores to determine the fakeness of a given video. For instance,

VT and AT are the results of intra-modality reasoning, and IM is the result of inter-modality reasoning for a given video of length N . Subsequently, we extract the indices $Indx$ with fake labels and calculate their probabilities individually based on VT , AT , and IM using Equations (7) and (8):

$$Indx [VT, AT, IM] = VT_i, AT_i, IM_i \forall i \in fake \quad (7)$$

$$Prob [VT, AT, IM] = \sum_{i=1}^N \left[\frac{VT_i}{N}, \frac{AT_i}{N}, \frac{IM_i}{N} \right] \forall i \in fake, \quad (8)$$

where $Indx$ holds the indices for fake labels of each reasoning module, and $Prob$ represents the probability of fake labels for each reasoning module. These probabilities were calculated based on the ratio of unusual (fake) emotion transitions for intra-modality reasoning and mismatched (fake) emotions for inter-modality reasoning over the total length N of a given video. For instance, if the $Prob$ of fake visual transitions VT exceeds 10% in a video of length N , then the visual modality will be considered tampered or deepfake. Subsequently, the selected indices and probabilities are used as tokens to generate textual explanations from a set of predefined vocabulary that can identify the localization of fake timestamps and specific modalities that are more understandable to non-AI experts as compared with the interpretability of Grad-CAM [6].

4 EXPERIMENTAL EVALUATION

This section presents the details of the experimental setup, optimal settings for the proposed method, dataset description, and performance analysis performed to demonstrate the effectiveness of the proposed neurosymbolic approach for DFD.

4.1 Experimental Setup and Dataset

To conduct the experiments, we converted the input videos into sets of cropped faces from the visual modality and the corresponding 0.5 s audio segments from the aural modality. The *FER* and *SER* models were trained on a system configured with a GPU with 24 GB of memory, 64 GB of RAM, and CUDA 11.0. The implementation was performed using the Tensorflow 2.0.5 and Keras 2.0.0 frameworks in Python 3.7, in the Anaconda environment. The proposed method was evaluated on the **Presidential Deepfakes Dataset (PDD)** [33], which contains 32 video clips of the two most recent US presidents, Donald Trump and Joseph Biden. Half of the videos were deepfake and were created using various deep learning techniques, such as Wav2Lip and fake audio. The complete dataset contained 20,425 frames, which were 11 min and 26 s in length. Unlike other methods that use a standard ratio of datasets for training and testing, we used the overall dataset presented in [33]. Figure 3(a) shows samples from the PDD. Additionally, we evaluated the proposed framework on the **World Leaders Dataset (WLD)** [34], which comprised videos featuring FaceSwap-manipulated images of various political leaders, including Obama, Hillary Clinton, Elizabeth Warren, and Bernie Sanders. Actual videos were obtained from YouTube, and the manipulated samples were created using swapped faces of the leaders' comedic impersonators. In this dataset, we used only actual (both visual and aural real) and deepfake (both visual and aural manipulated) samples and discarded the imposter class. Figure 3(b) shows some sample images from the WLD.

4.2 Evaluation of Emotion Detection Models

This section presents the performance evaluation of the visual and aural emotion detection models. For visual emotion detection, the *FER* model was tested on the FER2013 dataset [27], which comprised 35,887 samples of 48×48 pixel grayscale face images categorized into seven basic emotions. The *FER* model achieved an overall accuracy of 90.64%, and the confusion matrix is shown

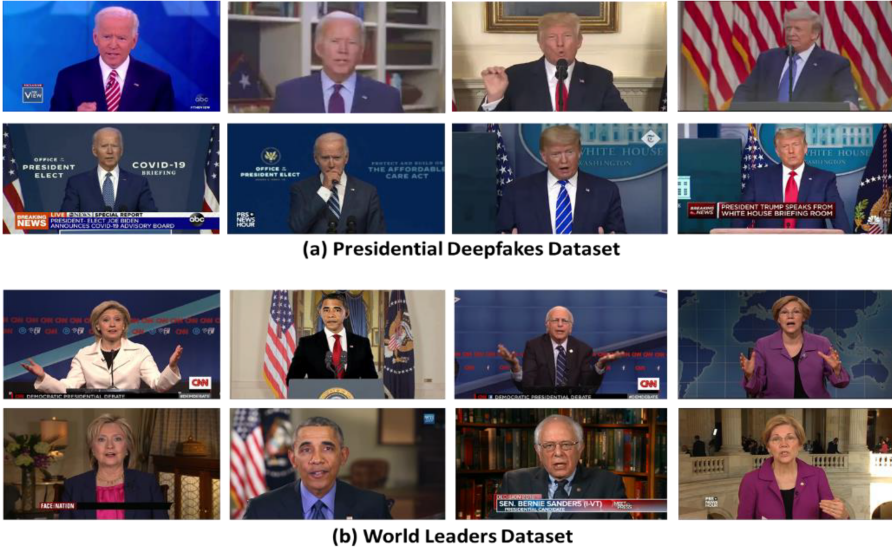


Fig. 3. Samples from (a) Presidential Deepfakes Dataset [33] and (b) World Leaders Dataset [34]. Top row: real; bottom row: fake.

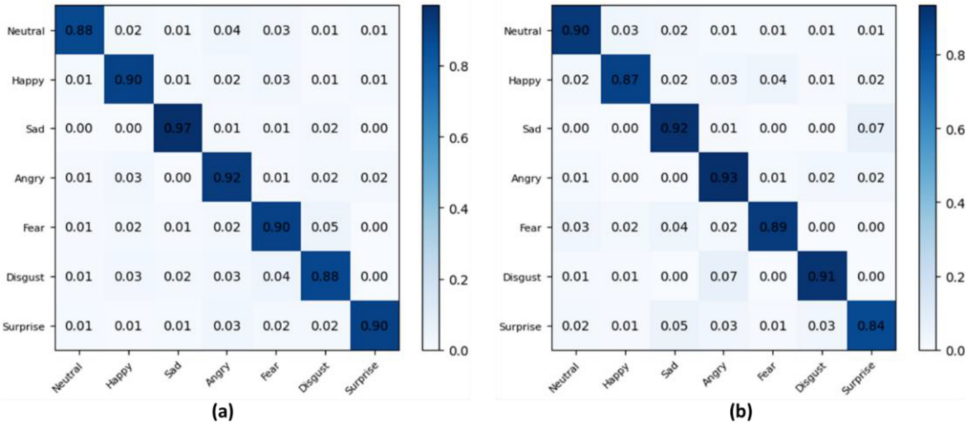


Fig. 4. Confusion matrix of (a) FER and (b) SER models.

in Figure 4(a). The *FER* model exhibited an accuracy exceeding 90% in recognizing each emotion except for neutral and disgust classes, where it exhibited an accuracy of 88% for those classes. Similarly, the *SER* model was tested on the RAVDESS dataset [28], which contained eight classes. In this study, we used seven basic emotions, where we excluded the “calm” class such that each modality has the same classes. In general, the *SER* model achieved an accuracy of 89.35%, and each class achieved an accuracy exceeding 84%, as shown in Figure 4(b).

4.3 Performance Evaluation on PDD

We evaluated the performance of the proposed framework by comparing it with the leading model from the **Deepfake Detection Challenge (DFDC)**, which was used in [34]. They provided a confidence score for each video prediction, which we similarly provided to evaluate the manner by

Table 3. Comparative Analysis of Proposed Neurosymbolic Approach with Approach Presented in [33]

Video	Ground Truth	DFDC [33]		Proposed (Intra-modality)	
		Confidence	Prediction	Confidence	Prediction
Biden 00	Fake	.63	Fake	.43	Fake
Biden 01		.96	Fake	.38	Fake
Biden 02		.96	Fake	.41	Fake
Biden 03		.50	–	.26	Fake
Biden 04		.96	Fake	.62	Fake
Biden 05		.97	Fake	.21	Fake
Biden 06		.97	Fake	.63	Fake
Biden 07		.95	Fake	.45	Fake
Trump 00		.27	Real	.31	Fake
Trump 01		.28	Real	.43	Fake
Trump 02		.20	Real	.39	Fake
Trump 03		.37	Real	.14	Real
Trump 04		.75	Fake	.19	Fake
Trump 05		.04	Real	.11	Real
Trump 06		.08	Real	.31	Fake
Trump 07		.79	Fake	.33	Fake
Biden 08	Real	.02	Real	.97	Real
Biden 09		.09	Real	.88	Real
Biden 10		.19	Real	.86	Real
Biden 11		.25	Real	.92	Real
Biden 12		.01	Real	.86	Real
Biden 13		.04	Real	.83	Fake
Biden 14		.85	Fake	.85	Real
Biden 15		.21	Real	.94	Real
Trump 08		.07	Real	.45	Fake
Trump 09		.27	Real	.88	Real
Trump 10		.57	Fake	.91	Real
Trump 11		.08	Real	.89	Real
Trump 12		.09	Real	.87	Real
Trump 13		.01	Real	.91	Real
Trump 14		.12	Real	.89	Real
Trump 15		.73	Fake	.87	Real
Overall Accuracy		69%		87.5%	

The Proposed Framework Correctly Classified Eight Videos That Were Incorrectly Classified in [33].

which the neurosymbolic approach addresses the misclassification of deep-learning-based solutions. For an unbiased comparison, we performed comparison with the intra-modality reasoning results of the visual modality. Table 3 presents a comparative analysis along with the confidence scores of the proposed framework and the results obtained from [33]. The DFDC model achieved an overall accuracy of 69% in the case where the model identified a likelihood exceeding 50% for the ground truth. In our framework, we set this threshold to 15%; i.e., if fake emotion transition in the video exceeds 15%, then the video is regarded as deepfake; otherwise, it is considered real. The proposed framework achieved an accuracy of 87.5% based on intra-modality reasoning (visual) without any prior training on the deepfake dataset. Moreover, Table 3 shows that the proposed

Table 4. Class-wise Comparative Analysis of Proposed Neurosymbolic Approach with GFGNN [36]

Method	Accuracy (%)			
	Obama	Clinton	Warren	Sanders
DFGNN [36]	69.1	65.8	67.6	63.4
Proposed	79.9	73.8	72.9	74.3

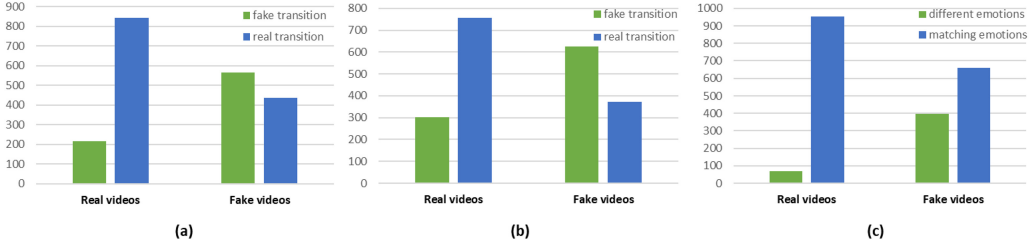


Fig. 5. Statistical analysis of Presidential Deepfakes Dataset based on proposed inter- and intra-modality reasoning: (a) intra-modality reasoning for visual, (b) intra-modality reasoning for aural, and (c) inter-modality reasoning for visual and aural.

framework correctly classified 8 among 10 predictions misclassified in [33]. Hence, the proposed neurosymbolic approach is more reliable and explainable than that proposed in [33].

4.4 Performance Evaluation on WLD

This section presents a performance evaluation of the proposed framework on the WLD [34]. The experimental analysis resulted in an overall accuracy of 75.34%. Although the overall accuracy was not extremely high, it was comparable with cross-dataset evaluations because the proposed framework was trained on the emotion dataset only. For instance, the recently developed method presented in [35] achieved an overall accuracy of 78.12% on the WLD when trained on the PDD. Similarly, the method proposed in [36] achieved accuracies of 69.36% and 76.1% for the WLD when trained on Celeb-DF and DFDC datasets, respectively. Table 4 presents the class-wise accuracy of the proposed method. The proposed method detects inconsistencies in computer-generated fake media; therefore, we discarded the imposter class from the evaluation. The Obama class for lip-sync and face swaps was regarded as one class and achieved a maximum accuracy of 79.9%. Similarly, Clinton, Warren, and Sanders achieved accuracies of 73.8%, 72.9%, and 74.3%, respectively.

4.5 Statistical Analysis of PDD

To further evaluate the proposed framework, we conducted a statistical analysis on the PDD to confirm our hypothesis and demonstrate the effectiveness of the neurosymbolic approach. As discussed in Section 3.2, we selected three frames and three overlapping audio segments per second for *FER* and *SER*. The total duration of videos in the PDD was 11 min and 26 s, which implies that approximately 2,058 sets of reasoning were performed for each modality. Figure 5 shows the statistics of the intra-modality reasoning for visual and aural fake emotion transitions and the inter-modality reasoning over the complete dataset. Clearly, the fake videos contained many fake emotion transitions. For instance, in the visual modality, only 20% of the fake emotion transitions were detected in real videos as compared with 56% in fake videos. Similarly, the ratio was 28:62 for the actual and fake videos under the aural modality. The results of the inter-modality reasoning can be similarly used to differentiate between matching and different emotions in real and fake

Table 5. Ablation Study of Proposed Neurosymbolic Approach

Modality	Intra-modality (visual)	Intra-modality (aural)	Inter-modality	Proposed (Inter + Intra)
Accuracy (%)	87.5	84.3	87.5	93.7

videos. The proposed inter-modality module detected 37.6% of different emotions in fake videos but only 6.9% of the same in real videos.

4.6 Ablation Study

We analyzed the effect of each reasoning module on the proposed framework, as listed in Table 5. For this analysis, we selected all 32 videos from the PDD and performed intra- and inter-modality reasoning individually. The proposed method achieved accuracy of 87.5% for the visual intra-modality and 84.3% for the aural intra-modality. Similarly, the accuracy for the inter-modality reasoning was 87.5%. However, when we combined all the results via voting, the proposed solution achieved an accuracy of 93.7%. Our voting strategy is simple: we select the final prediction if two of the three predictions are the same. Table 5 presents the results of the individual and combined analyses.

4.7 Discussion on Explainability and Generalizability

The development of explainable and generalizable methods is crucial for enhancing the trustworthiness and reliability of machine learning models. Recently, the field of explainable AI has advanced significantly owing to the emergence of new techniques and methodologies to improve the interpretability and comprehensibility of machine learning models. However, they are primarily intended for general purposes and focus on explaining the learning of deep models (i.e., Grad-CAM [6], LRP [37], and PRM [38]), which are not directly beneficial to non-AI experts. However, obtaining a generalized solution for all types of deepfakes is challenging because of the continuous evolution of deepfake generation algorithms. Thus, in this study, we aim to improve the explainability and generalizability of deepfake detection algorithms. One of the main advantages of the proposed framework is its ability to provide clear and concise explanations regarding the decisions made by the model using psychological knowledge of emotions. Our framework is based on a novel biological signal that combines deep neural networks with knowledge-based assisted symbolic reasoning, thus enabling both accurate predictions and interpretable explanations. Symbolic reasoning serves as an intuitive approach for representing the decision-making process of a framework, thus allowing it to be easily understood and interpreted. An example that explains real and fake videos is illustrated in Figure 6, which allows non-AI experts to easily understand the rationale behind the proposed framework's predictions. The *Indx* and *Prob* values from Equations (7) and (8) are used to visualize Figure 6, which indicate the timestamp at which the fake emotion transition and mismatch in aural and visual emotions occurred. For instance, Figure 6(a) shows an example of a real video classified as real, with extremely few fake emotion transitions and mismatched emotions (red highlighted portions). By contrast, a fake video classified as fake contains more fake emotion transitions and mismatched emotions in both modalities, with more red highlighted portions. In addition to explainability, the proposed framework exhibits a high degree of generalizability owing to the use of common knowledge and is independent of training on deepfake data. Our strategy enables the model to leverage knowledge learned from psychology and emotions to deepen fake domains. Thus, transferring common knowledge across domains renders the proposed framework more generalizable.

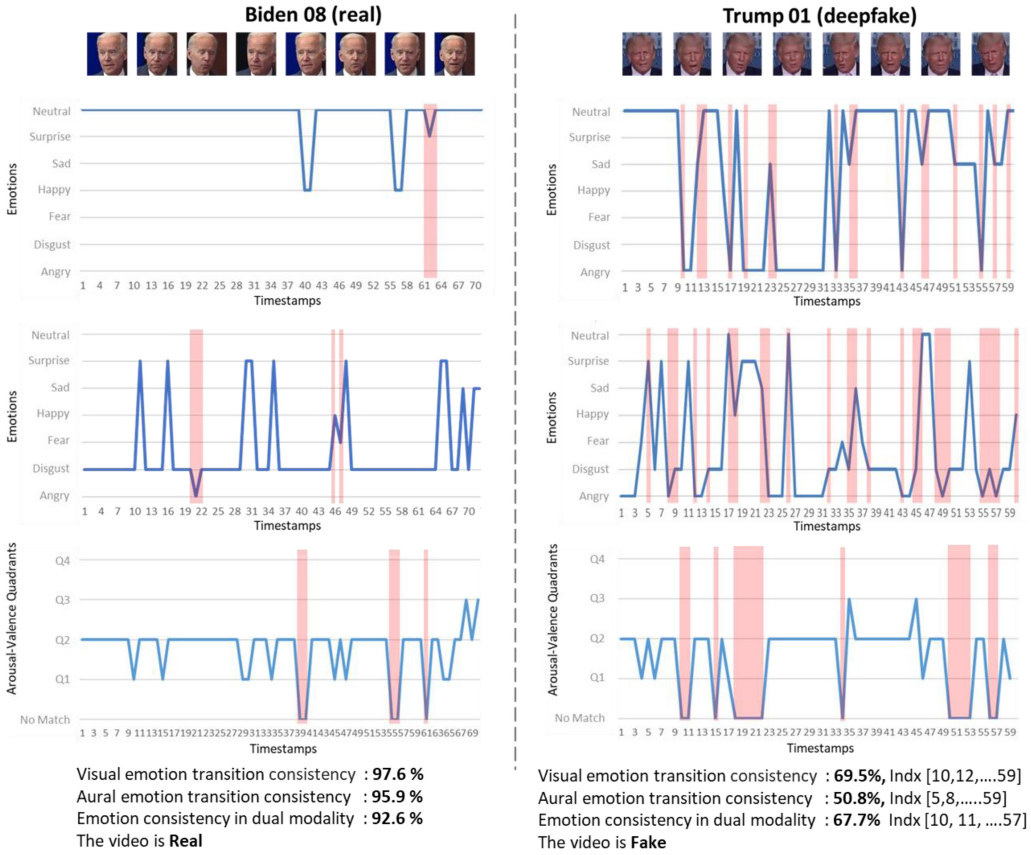


Fig. 6. Visual example of real and fake emotion transitions in real and deepfake videos, respectively.

5 CONCLUSION

In this study, a novel framework that combines psychological knowledge and symbolic reasoning to address the deepfake detection was proposed. The aim of this study was to improve the explainability and generalizability of deepfake detection methods. Hence, we introduced a novel biological signal (which demonstrates consistency between emotion transitions) based on a psychological analysis of intra-modality reasoning, along with arousal-valence dimensions for inter-modality reasoning. The proposed framework utilizes psychological knowledge, which can be explained in terms of emotional consistency. Furthermore, no deepfake dataset is required for training, which renders the proposed framework more generalizable owing to the underlying knowledge learned from psychology and emotions in deepfake domains. Thus, transferring common knowledge across domains renders the proposed framework more generalizable. We demonstrated that the proposed framework correctly classifies videos misclassified by the deep learning method, which confirmed the effectiveness of our approach in detecting deepfakes and highlighted the potential of a neurosymbolic approach for achieving explainability. The proposed framework demonstrated remarkable performance on the PDD, with an accuracy of 93.7%. Similarly, it achieved an overall accuracy of 75.34% on the WLD. In the future, we plan to introduce artifact cues along with biological signals to further improve performance over a diverse range of deepfake datasets.

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Received 30 March 2023; revised 31 July 2023; accepted 27 August 2023